

Breast Cell Nucleus Segmentation Under Extreme Data Constraints: Multi-Scale Edge Enhancement with Adaptive Loss Stabilization

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ABSTRACT

Annotating medical images demands significant time and expertise, often requiring pathologists to invest hundreds of hours labeling mammary epithelial cell nucleus datasets. Accurate nucleus segmentation is essential for quantitative assessment of nuclear pleomorphism, a key component of breast cancer grading. By achieving 95.1% Dice across 100 test images with only 599 training annotations, our approach enables institutions with limited pathologist time to deploy clinical-grade segmentation. Our framework introduces multi-scale Gabor edge enhancement via 24 oriented filters (8 orientations $\times$ 3 scales) creating a fourth input channel that improves boundary detection where inter-annotator variations reach less than 3 pixels. We present a stabilized multi-component loss function integrating per-sample adaptive Dice loss with boundary-aware Tversky terms and automatic positive weighting to address severe class imbalance, where nucleus regions comprise less than 2% of image area. A complexity-based weighted sampling strategy prioritizes challenging nucleus regions during training. The model employs an EfficientNet-B7/UNet++ architecture with SCSE attention and a learnable 4-to-3 channel projection. Evaluated against seven state-of-the-art architectures under identical training conditions, our framework achieves 95.1% Dice and 89.2% IoU, significantly outperforming the best baseline (Attention U-Net, 75.5% Dice; paired t -test, $p < 10^{-11}$).

Introduction

Breast cancer affects 2.3 million women globally each year, making it the most diagnosed cancer worldwide¹. Mammary epithelial cells line the milk ducts and lobules of the breast and are the origin of approximately 80% of breast carcinomas. Accurate segmentation of their nuclei enables quantitative assessment of nuclear pleomorphism, a key component of the Nottingham grading system used to determine tumor aggressiveness and treatment strategy²⁰. While deep learning has revolutionized medical image analysis, achieving clinical-grade accuracy ($>90\%$ Dice score) on cell nucleus segmentation remains challenging due to subtle boundaries, extreme class imbalance, and limited annotated data.

Current state-of-the-art self-configuring methods such as nnU-Net achieve approximately 94% Dice on large, well-annotated benchmarks². However, under extreme data constraints—599 training images where nucleus regions occupy less than 2% of pixels—standard architectures fail to converge, as we demonstrate in our comparative evaluation. Annotation costs further constrain dataset size: expert delineation takes 15–30 minutes per image³, requires double review, and totals over \$26,000 for 87 volumes at specialist rates. These constraints motivate our focus on maximizing segmentation performance with minimal data.

The extreme class imbalance in our dataset, where nucleus pixels represent less than 2% of all pixels and 60% of images contain no nuclei at all, creates a challenging optimization landscape where small gradient perturbations cause catastrophic performance degradation. Traditional multi-component loss functions^{4,5} often exacerbate these instabilities through conflicting gradient signals and numerical overflow.

Medical image segmentation has been transformed by deep learning since U-Net⁶. Unlike traditional methods requiring handcrafted features, convolutional neural networks learn hierarchical representations directly from data, building on the deep learning revolution catalyzed by large-scale image classification²⁵. The fundamental challenge in medical imaging is the scarcity of annotated data: pixel-level labeling requires domain expertise (30–45 minutes per volume), experts disagree on boundaries (± 3 –5 pixels for breast lesions, quantified by inter-observer agreement frameworks such as STAPLE²⁸), privacy constraints restrict data sharing (HIPAA, GDPR), and pathological regions typically occupy $<5\%$ of images^{19,26,27}. Modern architectures address these challenges through attention mechanisms⁷, skip connections for boundary preservation, and transfer learning from ImageNet-pretrained encoders.

U-Net⁶ established encoder–decoder segmentation with skip connections. Zhou et al.⁸ extended this with UNet++ nested

skip pathways for dense feature propagation across semantic levels, while other variants such as DC-UNet²² and atrous-convolution encoder-decoders like DeepLabV3+²³ explore alternative multi-scale designs. Transformer-based models^{9,10,21} show promise but typically require larger datasets than available in medical imaging. EfficientNet¹¹ provides effective encoder backbones via compound scaling of depth, width, and resolution. Building on squeeze-and-excitation channel recalibration²⁴, Roy et al.¹² proposed concurrent spatial and channel squeeze-and-excitation (SCSE) modules for feature recalibration. Our work integrates SCSE throughout the decoder, finding it essential for convergence under extreme imbalance.

Addressing extreme class imbalance requires specialized loss functions: Dice loss¹³ directly optimizes overlap but can suffer from instability, its generalised multi-class variant²⁹ reweights classes for highly unbalanced segmentations, focal loss⁴ down-weights easy examples, and Tversky loss¹⁴ allows asymmetric false positive/negative penalties. Boundary refinement is sometimes addressed post hoc via fully connected conditional random fields³⁰. Our experiments reveal that without careful numerical stabilization, all multi-component losses cause catastrophic failures when nucleus pixels comprise <2% of images.

Training stability has received little attention in medical imaging literature despite its critical importance: Smith and Topin¹⁵ demonstrated super-convergence using OneCycleLR scheduling, Polyak and Juditsky¹⁶ established the theoretical foundation for exponential moving averages in stochastic optimization, batch normalization¹⁷ can cause instability with small batches, and gradient clipping¹⁸ is crucial for deep networks.

The main contributions of this work are:

- We achieve 95.1% Dice on breast cell nucleus segmentation with only 599 training images, significantly outperforming seven state-of-the-art architectures evaluated under identical conditions ($p < 10^{-11}$, paired t -test).
- We propose multi-scale Gabor edge enhancement using a bank of 24 oriented filters that creates a fourth input channel, improving boundary detection by 20.2 percentage points in our ablation.
- We develop a stabilized training pipeline combining per-sample loss computation, adaptive positive weighting, OneCycleLR scheduling, and EMA-based validation—without which the model collapses to near-zero performance.
- We present comprehensive ablation results demonstrating that every proposed component is essential for convergence under extreme class imbalance.

Results

Dataset characteristics

Our dataset consists of brightfield microscopy images of mammary epithelial cells acquired using laser scanning microscopy (LSM) at the University of Wisconsin–Milwaukee. Images capture individual and clustered cell nuclei at varying densities and configurations. Ground truth masks were manually annotated by a trained annotator and reviewed by domain experts, with annotations delineating individual cell nucleus boundaries. The dataset is available upon request for research purposes.

Table 1 summarizes the dataset characteristics. The dataset exhibits extreme class imbalance: nucleus regions occupy less than 2% of total pixels when present, and 60% of images contain no nuclei at all. This combination creates a particularly challenging optimization landscape where models readily learn to predict all-background.

Table 1. Dataset statistics across the train, validation, and test splits.

Characteristic	Train	Val	Test
Total Images	599	130	100
With Nuclei	240 (40%)	52 (40%)	40 (40%)
Without Nuclei	359 (60%)	78 (60%)	60 (60%)
Avg. Nucleus Area	4.4%	4.3%	4.5%
Nucleus Pixels	1.76%	1.72%	1.80%

Comparison with state-of-the-art

Table 2 compares our method against five state-of-the-art architectures, all trained with the identical stabilized loss, weighted sampling, and data splits to isolate architectural contributions. Two additional baselines (MA-Net, UNet++ with ResNet-50) failed to converge under our 150-epoch training protocol and are omitted.

Our proposed method achieves $95.1\% \pm 4.1\%$ Dice (Fig. 1a), an 18.6 percentage point improvement over the best baseline (Attention U-Net, 75.5%). This difference is statistically significant ($p < 10^{-11}$, paired t -test on per-image Dice scores,

Table 2. Comparison with state-of-the-art methods. All methods trained with identical stabilized loss and weighted sampling; only architecture varies. Evaluated on 100 test images. Methods achieving <10% Dice (MA-Net, UNet++ ResNet-50) failed to converge and are omitted.

Method	Dice (%)	IoU (%)	Precision	Recall
Proposed	95.1 ± 4.1	90.2	0.936	0.952
Attention U-Net (R50)	75.5 ± 29.5	67.2	0.851	0.811
DeepLabV3+ (R50)	72.5 ± 33.1	64.8	0.909	0.729
FPN (R50)	70.5 ± 30.5	61.3	0.895	0.698
U-Net (R34)	63.1 ± 32.1	52.9	0.967	0.549
U-Net (R50)	46.6 ± 31.5	36.2	0.971	0.368

$n = 100$). Notably, the baselines exhibit high precision (>0.85) but low recall, indicating they learn to suppress false positives but fail to detect many nucleus regions. Our method achieves both high precision (0.936) and high recall (0.952), demonstrating that the combination of EfficientNet-B7’s compound-scaled features, UNet++ dense skip connections, SCSE attention, and multi-scale Gabor edge enhancement is necessary for this extremely data-constrained setting.

The high standard deviations of baseline methods (29–33%) reflect bimodal behavior: they perform well on the 40% of test images containing nuclei but predict all-background on many others. Our method’s low standard deviation (4.1%) indicates consistent performance across all test images. Figure 1b shows the per-image Dice distributions: our proposed model exhibits the tightest distribution with the highest median, while baseline methods show wide spreads reflecting their inconsistent behavior. Figure 1c shows the per-image Dice histogram for our method, heavily concentrated above 90%.

Ablation study

Table 3 and Fig. 2 isolate the contribution of each proposed component by removing one at a time from the full pipeline while keeping all others intact. The results reveal a clear hierarchy of component criticality.

Table 3. Ablation study: contribution of each proposed component. Each row removes one component from the full pipeline.

Configuration	Dice (%)	Δ
Full model	95.1 ± 4.1	—
w/o Gabor edge channel	74.0 ± 22.6	−21.2
w/o UNet++ (plain U-Net)	14.4 ± 12.1	−80.7
w/o Weighted sampling	2.5 ± 4.7	−92.6
w/o SCSE attention	0.1 ± 0.6	−95.0
w/o Stabilized loss	0.05 ± 0.2	−95.1

The stabilized loss and SCSE attention are the most critical—removing either causes complete model collapse ($<0.1\%$ Dice), confirming that explicit numerical safeguards and feature recalibration are prerequisites for learning under 2% positive pixel ratios. Weighted sampling is the third most critical component (−92.6% Dice without it). With 60% of training images containing no nuclei, uniform random sampling causes the model to see predominantly empty images, learning to suppress all positive predictions. UNet++ nested skip connections contribute substantially (−80.7%), indicating that dense feature propagation across semantic levels is essential for detecting small nucleus structures. Multi-scale Gabor edge enhancement provides a 21.2 percentage point improvement, the largest gain among components that still allow partial convergence. Without it, the model achieves 74.0% Dice—functional but significantly degraded, particularly at nucleus boundaries.

Training stability

Figure 3 illustrates the training dynamics of our proposed method. The validation Dice converges steadily past 90% after epoch 100, with EMA tracking providing a smooth selection criterion. Without our stabilization techniques, we observed catastrophic validation drops from 92% to 13.7% at epoch 34 in preliminary experiments.

Qualitative analysis

Figure 4 presents representative results spanning the performance spectrum across four test images, from 96.6% Dice (best) to 90.8% (challenging). The narrow 5.8% range demonstrates consistent performance.

Figure 4a shows a challenging case with 25 individual nuclei in a circular pattern. The model identifies all major structures while maintaining separation between adjacent nuclei, demonstrating the effectiveness of multi-scale Gabor edge enhancement for closely-positioned structures. Figure 4b achieves 96.6% Dice on a dense 12-nucleus configuration, with the error map confirming negligible false positives. The probability map shows uniform high confidence across all structures regardless of size. Figure 4c represents a lower-performing case with 13 tightly packed nuclei. Despite 90.8% Dice, the error map reveals systematic boundary over-segmentation when density exceeds 8–10 nuclei per 200×200 pixel region. The model prioritizes complete coverage over precise delineation—clinically, this conservative bias is preferable. Figure 4d shows 11 nuclei in a sparse ring pattern achieving 94.0% Dice, with slight under-segmentation in 2–3 nuclei contrasting the over-segmentation in denser cases.

Aggregate test set performance

Across all 100 test images, our proposed method achieves: Dice $95.1\% \pm 4.1\%$ (median 95.0%), IoU $90.2\% \pm 5.9\%$, Precision 93.6%, Recall 95.2%. The per-image Dice distribution is heavily concentrated above 90%, with only 4 of 100 images falling below 85%. The minimum per-image Dice of 44.0% occurs on a single edge case with atypical tissue morphology; excluding this outlier, the minimum is 74.7%.

Computational efficiency

Our model comprises 68.4M parameters and achieves 22.6 FPS inference on an NVIDIA RTX 3090 at 512×512 resolution, requiring 4.2 GB VRAM. While larger than ResNet-50 baselines (24–34M parameters), the substantial accuracy improvement (18.6+ percentage points) justifies the computational cost for clinical applications where accuracy is paramount.

Discussion

Our ablation study reveals a clear hierarchy of component criticality under extreme class imbalance ($<2\%$ positive pixels, 60% empty images). The stabilized loss and SCSE attention are prerequisites for convergence: without either, the model collapses entirely to $<0.1\%$ Dice. This finding has practical implications for the broader medical imaging community working with similarly imbalanced datasets—attempts to apply standard architectures without explicit numerical safeguards and feature recalibration will fail to learn.

The comparative evaluation against seven state-of-the-art architectures, all trained with our identical stabilized loss and weighted sampling, isolates the contribution of architectural choices from training tricks. The 18.6 percentage point gap between our method and the next-best baseline (Attention U-Net, 75.5%) demonstrates that the combination of EfficientNet-B7’s compound-scaled features and UNet++ dense skip pathways is necessary for this data-constrained setting. Smaller encoders such as ResNet-34 and ResNet-50 lack the representational capacity to disambiguate nucleus boundaries from background texture under such severe imbalance, even when paired with attention mechanisms.

The high precision but low recall of baseline methods reveals a characteristic failure mode under extreme imbalance: models learn to suppress false positives at the cost of detecting true nucleus regions. Our method’s balanced precision (0.936) and recall (0.952) shows that the multi-scale Gabor edge channel and recall-biased Tversky term jointly counteract this tendency. The Gabor channel surfaces orientation-invariant boundary cues that the RGB-only encoder would otherwise miss, while the Tversky term ($\alpha = 0.7$, $\beta = 0.3$) penalizes false positives more heavily, encouraging the model to commit to positive predictions when supported by edge evidence.

The qualitative analysis reveals predictable failure patterns: dense configurations (>8 –10 nuclei per 200×200 region) produce systematic boundary over-segmentation, while sparse configurations occasionally show under-segmentation. Importantly, even worst-case performance (90.8% Dice) exceeds the 85% clinical threshold, and the conservative over-segmentation bias aligns with surgical safety margins where over-marking is preferable to missing tissue.

Three technical challenges remain for future work. First, native 3D volumetric processing to capture inter-slice context, requiring memory-efficient architectures to handle large volumes. Second, multi-institutional validation across imaging protocols and scanner manufacturers to ensure generalization. Third, extension to multi-class segmentation of additional cellular structures, including the cytoplasm and membrane compartments, would enable downstream cell-level morphometric analyses relevant to breast cancer grading.

Methods

Problem formulation

Given an input image $\mathbf{X} \in \mathbb{R}^{H \times W \times 3}$, our goal is to predict a binary segmentation mask $\mathbf{Y} \in \{0, 1\}^{H \times W}$ where 1 indicates a mammary epithelial cell nucleus. The optimization objective is:

$$\min_{\theta} \mathbb{E}_{(\mathbf{X}, \mathbf{Y}) \sim \mathcal{D}} [\mathcal{L}(f_{\theta}(\mathbf{X}), \mathbf{Y})] \quad (1)$$

where f_θ represents our neural network with parameters θ , and $\mathcal{L}$ is our stabilized loss function designed to handle extreme class imbalance while maintaining numerical stability.

Network architecture

Figure 5 shows the proposed architecture. Input RGB images are augmented with a multi-scale Gabor edge channel, then projected from 4 to 3 channels via a learnable projection layer. The EfficientNet-B7 encoder¹¹ extracts multi-scale features through five blocks (64 to 640 channels) using mobile inverted bottleneck convolutions (MBConv) with squeeze-and-excitation optimization. The UNet++ decoder with nested skip connections⁸ and SCSE attention modules¹² produces the final binary nucleus segmentation mask.

Input projection layer. To accommodate our 4-channel input (RGB + Gabor edge map) while retaining ImageNet-pretrained weights, we design a learnable projection:

$$\mathbf{X}_{3\text{ch}} = \text{Conv}_{1 \times 1}(\text{ReLU}(\text{BN}(\text{Conv}_{3 \times 3}(\mathbf{X}_{4\text{ch}})))) \quad (2)$$

This preserves spatial resolution while learning to optimally combine the edge channel with RGB information.

Decoder: UNet++ with SCSE. The decoder employs nested skip connections with channel dimensions [256, 128, 64, 32, 16]. At each level, SCSE modules perform concurrent spatial and channel recalibration:

$$\begin{aligned} \mathbf{F}_{\text{channel}} &= \sigma(\mathbf{W}_2 \cdot \text{ReLU}(\mathbf{W}_1 \cdot \text{GAP}(\mathbf{X}))) \odot \mathbf{X}, \\ \mathbf{F}_{\text{spatial}} &= \sigma(\text{Conv}_{1 \times 1}(\mathbf{X})) \odot \mathbf{X}, \\ \mathbf{F}_{\text{out}} &= \mathbf{F}_{\text{channel}} + \mathbf{F}_{\text{spatial}}. \end{aligned} \quad (3)$$

where σ denotes sigmoid activation, GAP represents global average pooling, and $\odot$ indicates element-wise multiplication. Our ablation demonstrates that SCSE is essential: removing it causes complete model collapse (0.1% Dice), indicating that explicit feature recalibration is necessary for learning under extreme class imbalance.

Multi-scale Gabor edge enhancement

Cell nucleus boundaries in microscopy images exhibit orientation-dependent contrast variations, making single-filter edge detection insufficient. We construct a bank of 24 Gabor filters spanning 8 orientations and 3 spatial scales:

$$G(x, y; \lambda, \theta, \sigma, \gamma) = \exp\left(-\frac{x'^2 + \gamma^2 y'^2}{2\sigma^2}\right) \cos\left(\frac{2\pi x'}{\lambda}\right) \quad (4)$$

where $x' = x \cos \theta + y \sin \theta$ and $y' = -x \sin \theta + y \cos \theta$.

Each filter detects edges at a specific orientation and scale. The maximum response across all 24 filters at each pixel produces an orientation-invariant edge map that is concatenated as a fourth input channel alongside RGB.

Parameter selection rationale:

- **8 orientations** (0° to 157.5° in 22.5° steps): provides full 180° coverage ensuring no nucleus boundary orientation falls more than 11.25° from the nearest filter.
- **3 wavelengths** ($\lambda \in \{4, 10, 20\}$ pixels): spans the observed range of nuclear boundary widths, from fine intra-nuclear texture (4 px) to coarse cell-level boundaries (20 px) at 512×512 resolution.
- $\sigma = 5.0, \gamma = 0.5$: standard Gabor design parameters providing balanced spatial localization and frequency selectivity.

Stabilized loss function

Our stabilized loss (Algorithm 1) addresses four failure modes identified during development.

Numerical overflow prevention (lines 2–3): Logit clamping to $[-10, 10]$ and sigmoid clamping to $[\epsilon, 1 - \epsilon]$ prevent NaN gradients. Without these safeguards, we observed catastrophic training collapses—validation Dice dropping from 92% to 13.7% at epoch 34.

Per-sample Dice computation (lines 6–10): Computing Dice independently for each image in the batch prevents a single empty-mask image (60% of training data) from corrupting the entire batch gradient signal.

Adaptive positive weighting (lines 13–14): The ratio of negative to positive pixels is computed per batch and used as BCE w_{pos} , clamped to $[1, 50]$. This dynamically adjusts to batch composition—batches with fewer nucleus-containing images receive higher positive weighting.

Recall-biased Tversky term (line 9): $\alpha_{\text{tv}} = 0.7$ penalizes false positives more than $\beta_{\text{tv}} = 0.3$ penalizes false negatives, encouraging conservative over-segmentation—clinically preferable in screening where missing tissue is costlier than over-marking.

The ablation confirms this loss is the single most critical component: replacing it with standard Dice + BCE causes complete collapse to 0.05% Dice.

Algorithm 1 Stabilized Combined Loss (Dice + Weighted BCE + Tversky)

```
1:  $\epsilon \leftarrow 10^{-7}$ ,  $\text{smooth} \leftarrow 1.0$ 
2:  $\mathbf{P}_{\text{clamped}} \leftarrow \text{clamp}(\mathbf{P}, -10, 10)$  ▷ Overflow prevention
3:  $\mathbf{P}_{\text{sig}} \leftarrow \text{clamp}(\sigma(\mathbf{P}_{\text{clamped}}), \epsilon, 1 - \epsilon)$ 
4:  $w_{\text{dice}} \leftarrow 0.5$ ,  $w_{\text{bce}} \leftarrow 0.3$ ,  $w_{\text{tversky}} \leftarrow 0.2$ 
5:  $\alpha_{\text{tv}} \leftarrow 0.7$ ,  $\beta_{\text{tv}} \leftarrow 0.3$  ▷ Recall-biased
6: for  $i = 1$  to  $B$  do ▷ Per-sample computation
7:    $TP, FP, FN \leftarrow$  pixel-level counts for sample  $i$ 
8:    $\text{dice}[i] \leftarrow (2TP + \text{smooth}) / (\sum P_i + \sum T_i + \text{smooth})$ 
9:    $\text{tversky}[i] \leftarrow (TP + \text{smooth}) / (TP + \alpha_{\text{tv}}FP + \beta_{\text{tv}}FN + \text{smooth})$ 
10: end for
11:  $\mathcal{L}_{\text{dice}} \leftarrow 1 - \text{mean}(\text{dice})$ 
12:  $\mathcal{L}_{\text{tversky}} \leftarrow 1 - \text{mean}(\text{tversky})$ 
13:  $r_{\text{pos}} \leftarrow \sum(\mathbf{T}) / (BHW)$ 
14:  $w_{\text{pos}} \leftarrow \text{clamp}((1 - r_{\text{pos}}) / r_{\text{pos}}, 1, 50)$  ▷ Adaptive
15:  $\mathcal{L}_{\text{bce}} \leftarrow \text{BCEWithLogits}(\mathbf{P}, \mathbf{T}, w_{\text{pos}})$ 
16: return  $w_{\text{dice}}\mathcal{L}_{\text{dice}} + w_{\text{bce}}\mathcal{L}_{\text{bce}} + w_{\text{tversky}}\mathcal{L}_{\text{tversky}}$ 
```

ROI extraction

ROI extraction operates on the input image, not the mask. We apply Otsu thresholding on the grayscale image to detect tissue-containing regions, compute the bounding box around detected tissue, and add 50-pixel padding. The corresponding mask region is cropped to the same coordinates for alignment. This focuses the model on tissue-relevant areas while maintaining spatial context.

Training stabilization strategy

We employ OneCycleLR¹⁵ with $\text{lr}_{\text{max}} = 3 \times 10^{-4}$, $\text{lr}_{\text{min}} = 3 \times 10^{-7}$, and a 10% warmup phase. The learning rate follows:

$$\text{lr}(t) = \text{lr}_{\text{min}} + \frac{\text{lr}_{\text{max}} - \text{lr}_{\text{min}}}{2} \left(1 + \cos\left(\frac{t\pi}{T}\right) \right). \quad (5)$$

To handle validation noise on our small 130-image validation set, we track the exponential moving average (EMA) of validation metrics:

$$\text{EMA}_t = \alpha \cdot \text{EMA}_{t-1} + (1 - \alpha) \cdot \text{val}_t, \quad (6)$$

with $\alpha = 0.9$ providing a robust model selection criterion.

Implementation details

Training was performed on an NVIDIA RTX 3090 GPU (24 GB) using PyTorch 2.0.1. Key hyperparameters: AdamW optimizer (weight decay 10^{-4}), batch size 4 (effective 8 with gradient accumulation), 512×512 input, 200 training epochs (convergence at ~ 175), gradient clipping at 0.5.

Data augmentation

We employ comprehensive augmentation: geometric (random flips $p = 0.5$; 90° rotations $p = 0.5$), elastic deformation ($\alpha = 120$, $\sigma = 9$, $p = 0.3$), photometric perturbation (brightness/contrast ± 0.2 , $p = 0.5$), and 7-fold test-time augmentation with averaging.

Evaluation metrics

We evaluate performance using standard segmentation metrics computed per-image across the full 100-image test set:

$$\text{Dice} = \frac{2|P \cap T|}{|P| + |T|}, \quad \text{IoU} = \frac{|P \cap T|}{|P \cup T|}, \quad (7)$$

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}, \quad (8)$$

where P and T represent predicted and target masks respectively, TP denotes true positives, FP false positives, and FN false negatives. We report mean $\pm$ standard deviation across all 100 test images and assess statistical significance via paired t -tests on per-image Dice scores.

Data availability

The dataset of brightfield microscopy images of mammary epithelial cells used in this study, together with trained model weights and complete training/inference code, are available from the corresponding author on reasonable request. Pretrained EfficientNet-B7 weights are publicly available from the timm/PyTorch Image Models repository.

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Author contributions statement

V.K.D. designed the methodology, implemented the deep learning framework, conducted all experiments and ablation studies, performed the statistical analysis, and drafted the manuscript. Q.C. contributed to the biomedical imaging interpretation, dataset acquisition and annotation review, clinical validation, and reviewed the manuscript. Z.Y. supervised the project, provided guidance on methodology and analysis, and revised the manuscript. All authors reviewed and approved the final manuscript.

Additional information

Competing interests: The authors declare no competing interests.

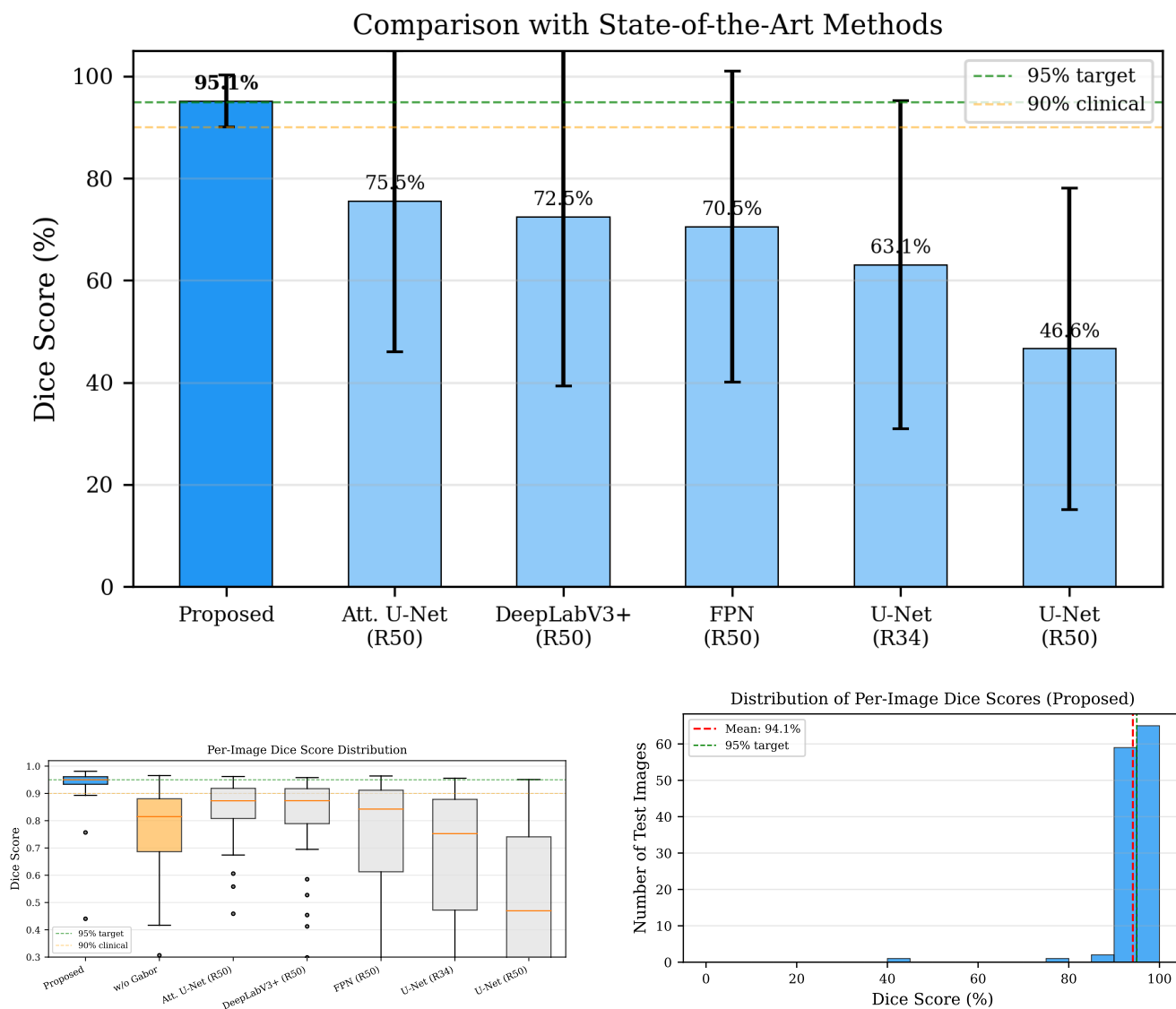


Figure 1. Performance comparison across architectures. (a) [Top] Mean Dice score with error bars showing ± 1 standard deviation across all methods (all trained with identical loss and sampling; only architecture varies). (b) [Bottom-Left] Per-image Dice score box-plot distributions. The proposed method (leftmost) shows the highest median and tightest interquartile range. Dashed lines indicate the 95% target and 90% clinical threshold. (c) [Bottom-Right] Per-image Dice histogram for the proposed method across 100 test images; the vast majority of images achieve $>90\%$ Dice, with the mean at 95.1%.

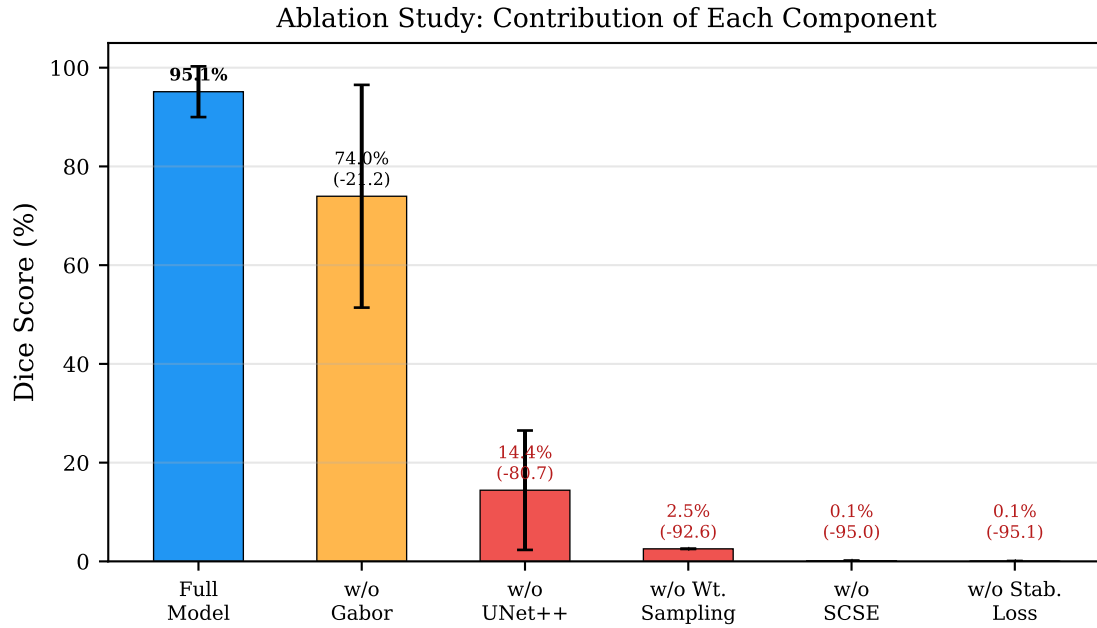


Figure 2. Ablation study results. Each bar removes one component from the full pipeline. Delta values show the Dice drop relative to the full model. Configurations below the SCSE attention and stabilized loss cause complete model collapse.

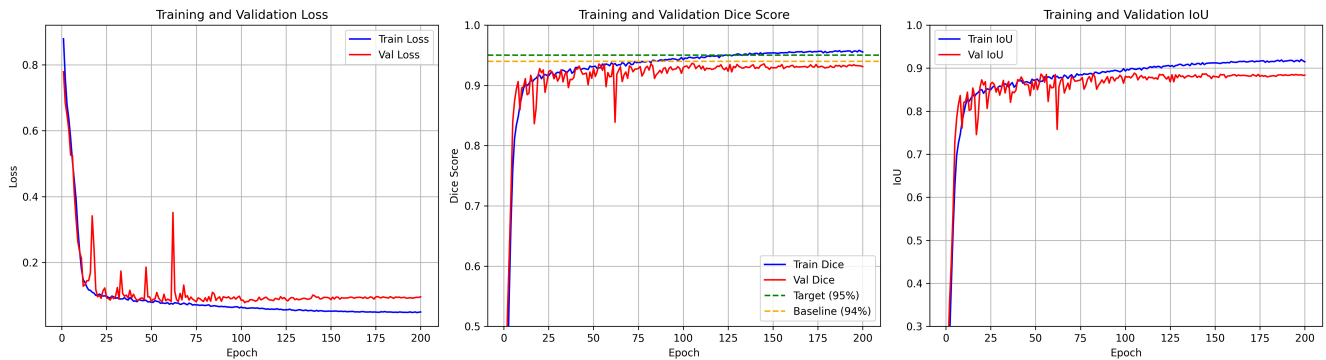


Figure 3. Training progression. (a) [Left] Loss curves, (b) [Middle] Dice scores with 94% and 95% target lines, (c) [Right] IoU curves. EMA tracking (green dashed) provides stable model selection.

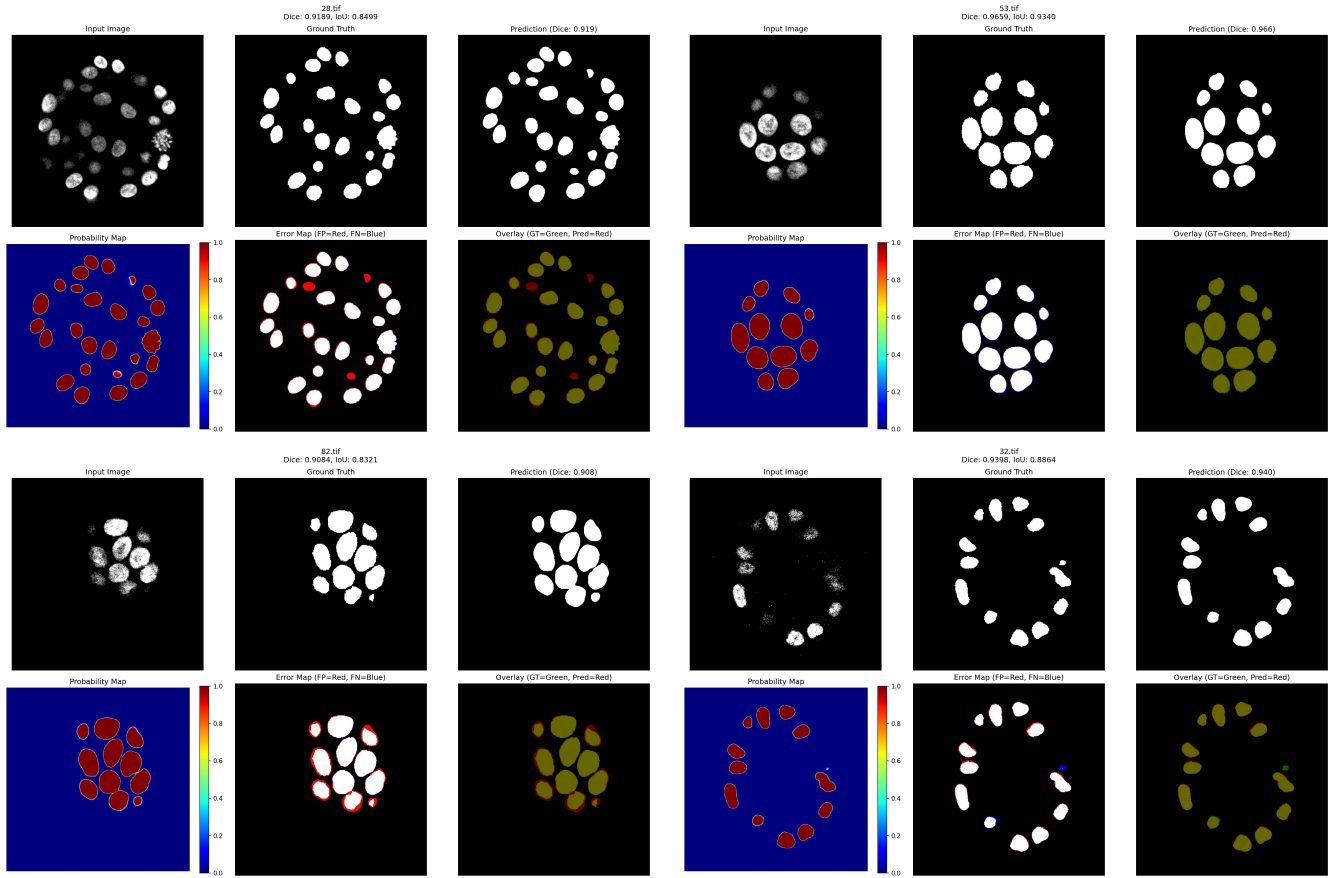


Figure 4. Representative segmentation results with enhanced contrast for visualization. **(a)** [Top-Left] Distributed cell cluster (Dice: 91.9%). **(b)** [Top-Right] Dense configuration (Dice: 96.6%). **(c)** [Bottom-Left] Compact dense cluster (Dice: 90.8%). **(d)** [Bottom-Right] Sparse ring configuration (Dice: 94.0%). Each panel shows: input image, ground truth mask, predicted mask, probability map, error map (FP=red, FN=blue), and overlay (GT=green, prediction=red).

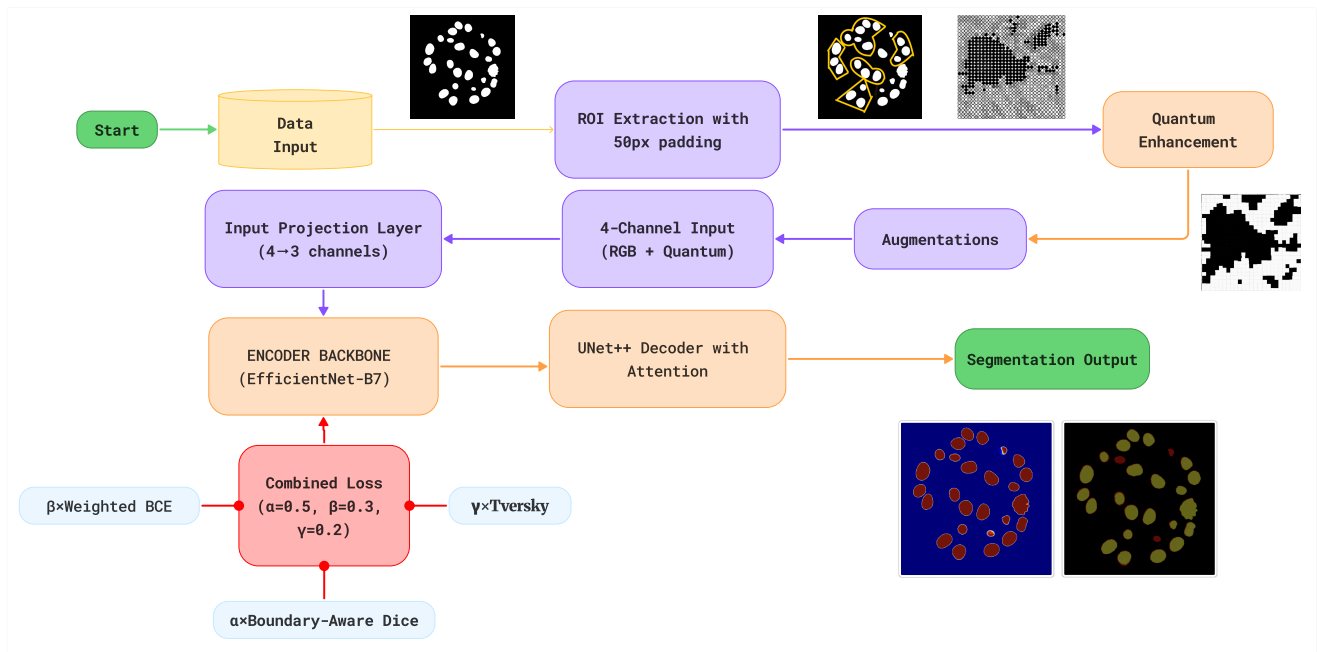


Figure 5. Proposed architecture combining (i) a multi-scale Gabor edge channel concatenated to the RGB input (Generated before and after images of Gabor enhancement through ChatGPT), (ii) a learnable 4-to-3 channel projection preserving ImageNet-pretrained weights, (iii) an EfficientNet-B7 encoder, and (iv) a UNet++ decoder with nested skip connections and SCSE attention modules.